

SAI POOJITHA SAJJAVARAPU

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Education

Rajiv Gandhi University of Knowledge and Technologies (IIIT Nuzvid)

Bachelor of Technology in Computer Science, **CGPA: 8.5**

Nuzvid, Eluru

September 2022 – April 2026

Experience

Quantum Gandiva AI

Onsite (November 2025 – Present)

AI Backend Engineer & Data Engineer

Visakhapatnam

- Built the **entire backend stack** for a production Agentic AI product, handling agent orchestration, execution logic, and system coordination across **7+ autonomous agents** and multi-tool workflows.
- Shipped **FastAPI backend services** with asynchronous request handling and modular APIs, **reducing average response latency by ~40%** while integrating LLM reasoning, agent memory, and dynamic tool-calling.
- Implemented **AWS-backed data pipelines** (S3, Lambda) for real-time ingestion and decision-making, **processing ~10K+ events/day** with low operational overhead.

Parabola9

Onsite (December 2024 – May 2025)

AI/ML Developer

Nuzvid, Eluru

- Developed and optimized advanced **machine learning models** to solve complex problems, including **generative AI applications** and **LLM-based systems**, **improving task accuracy by ~20%**.
- Deployed **AI models** in production with scalable integration and continuously monitored and refined performance, **cutting inference latency by ~30%** to ensure efficiency and adaptability.
- Collaborated with engineers and domain experts to integrate and **deploy models end-to-end**, improving system efficiency and reliability in production.

Projects

Tailor-Fit: AI-Powered Shopping Platform 🛍️

- Built an **AI fashion-tech platform** with an end-to-end **ML/CV pipeline** for body-measurement estimation and virtual try-on, serving scalable **real-time inference (~2s per request)**.
- Automated measurement extraction and tailoring workflows, achieving a **35% simulated return-rate reduction** and improved precision fitting via a connected local tailor network.
- Tech Stack:** React.js, Node.js, Express.js, Python, TensorFlow, PyTorch, PiHuD, HR-VITON

AI-Driven Blood Donation Network: Intelligent Blood Donor–Bank Matching System 🩸

- Developed an **AI-driven blood donation platform** integrating **FastAPI (backend)** and **Flutter (frontend)** for real-time donor–blood bank connectivity.
- Implemented **intelligent donor–bank matching** using **Vertex AI**, **LangChain**, and **FAISS vector search** to enhance accuracy and **reduce response latency by ~35%**.
- Integrated **Google Maps API** with geofencing and smart alerts for **location-aware donor mobilization** and emergency blood request handling.
- Tech Stack:** FastAPI, Flutter, Firebase, Vertex AI, LangChain, FAISS, Google Cloud

MediBuddy: AI-Powered Prescription Interpreter 🩺

- Developed a web application that interprets medical prescriptions and simplifies terminology using **Gemini LLM API via backend API integration**, focused on accessibility for elderly users.
- Designed a modular, responsive UI with **React.js** (Hooks, Context API) with **voice output for summaries** and error-resistant API handling.
- Tech Stack:** React.js, Tailwind CSS, Vite, **Gemini API**, PostCSS, ESLint, JavaScript, **Docker**

Technical Skills

- Programming Languages:** Java, C, C++, Python, JavaScript, Dart
- Technologies & Frameworks:** **FastAPI**, Node.js, Django, REST APIs, **Agentic AI**, LLMs, **Generative AI**, NLP, **Machine Learning**, **Deep Learning**, Flutter, React.js
- Tools & DevOps:** Git, Docker, Kubernetes, Postman, GitHub Actions, CI/CD, **Redis (Streams)**
- Databases & Cloud Platforms:** MySQL, PostgreSQL, AWS (EC2, S3, Lambda, RDS, VPC), GCP

Achievements

- Smart India Hackathon 2024 – National Winner:** Led a team to **1st Prize** at India's largest national hackathon by building an AI/ML solution to address **bus bunching** in public transportation.
- Teczite Mega Expo 1st Prize Winner:** Secured **1st Prize** at a national-level expo by developing an **AI-powered security surveillance system** integrating **gesture recognition**, **YOLO-based anomaly detection**, and **real-time video captioning**.
- IIIT Hackathon Winner – Parabola9:** Won **1st Prize** for developing **VideoGPT**, a deep learning-based tool for generating video captions using frame optimization.
- Problem-Solving Expertise:** Solved over **500+ LeetCode problems** with a max contest rating of **1600+**.